On Quantum Mechanics: A Sketch

Abstract

We derive the Schrödinger equation of quantum mechanics following the steps in stochastic process. As a sketch, detailed proofs are temporally omitted.

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1 How to Read this Note?

Tips for reading this note:

- **Definitions** are in bold font.
- *Important* statements are in italic font.
- Only important equations are numbered.
- Questions are in red color.
- Conclusions are in green color.
- You can omit all the nonsense about stochastic process, just focusing on quantum mechanics.
- Footnotes are also negligible (except for those containing **definitions**).

2 On Wave-Function

In classical physics, a system with n classical particles in 3-dimensional space can be represented by an element in $\mathbb{R}^{3n}$, or a **configuration**. The time evolution is thus a trajectory in the space of configurations, namely a map $\mathbb{R} \to \mathbb{R}^{3n}$.[1] Generally, we suppose that the space of configurations is d-dimensional Euclidean.

The state of a quantum system is represented by wave-function. A **wave-function** is a map from the space of configurations to complex plane, say $f: \mathbb{R}^d \to \mathbb{C}$. In traditional textures, wave-functions are supposed to be square-integrable, thus $f \in L^2(\mathbb{R}^d)$. This is essential for probability interpretation (axiom 2). It is well known, however, that the Fourier transform of a square-integrable function may not be square-integrable again, so that its inverse Fourier transform may not exist. Since Fourier transform is basic in quantum mechanics, we shall seek for a smaller space in which wave-functions live. An ideal substitution is the (complex) Schwartz space. A **Schwartz space** $\mathcal{S}(\mathbb{R}^d)$ contains smooth function $f: \mathbb{R}^d \to \mathbb{C}$ that decays "exponentially fast" at infinity. Precisely, for any m-order polynomial P_m and any m-order partial derivative D^n, with integers m, $n \geqslant 0$, we have

$$\lim_{\|x\| \to \infty} |P_m(x) D^n f(x)| = 0.$$

Functions in Schwartz space are usually termed as **Schwartz functions**. Fourier transform is an automorphism on Schwartz space. And for any $f \in L^2(\mathbb{R}^d)$ and any $\varepsilon > 0$, there is a $g \in \mathcal{S}(\mathbb{R}^d)$ such that[2]

$$\int_{\mathbb{R}^d} \mathrm{d}x\, |f(x) - g(x)|^2 < \varepsilon.$$

That is, $\mathcal{S}(\mathbb{R}^d)$ is a dense subspace in $L^2(\mathbb{R}^d)$. It is in this sense that the substitution is plausible.

Restriction from $L^2(\mathbb{R}^d)$ to $\mathcal{S}(\mathbb{R}^d)$, however, is still insufficient. For example, in solving the stationary Schrödinger equation of one-dimensional harmonic oscillator, we suppose that wave-function has the form

$$f(x) = \exp(-x^2) \left[\sum_{n=0}^{\infty} a_n x^n \right],$$

where a_ns are to be determined. The factor $\exp(-x^2)$ is employed for an exponentially fast decay as $|x| \to \infty$ (thus faster than the inverse of any polynomial). And the factor $[\cdots]$ is a Taylor series. Hence, f is an analytic function in Schwartz space. The energy quantization emerges for ensuring the convergence of the series $\sum_n a_n x^n$. It indicates that we shall further restrict the space of wave-function to **analytic Schwartz space**, denoted by $\mathcal{S}_A(\mathbb{R}^d)$, which collects all the analytic functions in Schwartz space $\mathcal{S}(\mathbb{R}^d)$. Analytic functions are dense in Schwartz space, meaning that for any $f \in \mathcal{S}(\mathbb{R}^d)$ and any $\varepsilon > 0$, there is a $g \in \mathcal{S}_A(\mathbb{R}^d)$ such that[3]

$$\sup_{x \in \mathbb{R}^d} |f(x) - g(x)| < \varepsilon.$$

Notice that the norms employed for quantifying the approximations are different. While considering $\mathcal{S}(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$, we used the norm in $L^2(\mathbb{R}^d)$, integral on the squared norm of the difference, but in $\mathcal{S}_A(\mathbb{R}^d) \subset \mathcal{S}(\mathbb{R}^d)$, we used the L_1-norm, the supremum on the norm of the difference.

To prove that analytic functions are dense in Schwartz space, we introduce the convolution for a function $f \in \mathcal{S}(\mathbb{R}^d)$ as

$$f_n(x) := \int_{\mathbb{R}^d} \mathrm{d}y \delta_n(x - y) f(y)$$

by a "kernel" function

$$\delta_n(x) := \left(\frac{n}{2\pi}\right)^{d/2} \exp\left(-\frac{n}{2}\|x\|^2\right),$$

1. Remark that the space of configurations is not the phase space. For a Hamiltonian system, the phase space is $\mathbb{R}^{6n}$, where the dimension of phase space doubles because of momenta.

2. TODO: give a reference or a proof.

3. TODO: give a reference or a proof.

where $n = \{1, 2, \ldots\}$. The δ_n function is recognized as Gaussian with variance $1/n$. This construction is fantastic, because derivatives taken on f_n are now converted onto δ_n instead of f, which is even not necessarily continuous! But, as a Schwartz function, f has been "good" enough; it is uniformly bounded on $\mathbb{R}^d$. Indeed, $f(x)$ is negligible when x is far from origin, so that we regard f as compact supported, then f is uniformly bounded because of its continuity. This also applies to the derivatives of f since they are also Schwartz functions. Then, analyticity raises from the good properties of δ_n. To show that $f_n(x)$ tends to $f(x)$ as n tends to infinity, we first change the variable of integral by replacing $y \to (x - y)$, thus $f_n(x) = \int \mathrm{d}y \delta_n(y) f(x - y)$. Then using $\int \mathrm{d}x \delta_n(y) = 1$, we find

$$f_n(x) - f(x) = \int_{\mathbb{R}^d} \mathrm{d}y \delta_n(y)[f(x - y) - f(x)].$$

Notice that δ_n is much sharper a peak at origin as n goes greater, together with the fact that ∂f is uniformly bounded, we find, for any $\varepsilon > 0$, there is an $N > 0$ such that $|f_n(x) - f(x)| < \varepsilon$ for any $n > N$ and $x \in \mathbb{R}^d$. In other words, we get $\lim_{n\to\infty} f_n = f$, thus proof ends. This is an elegant proof because we can *see* how f is approximated by a series of analytic functions explicitly.[4] As a by-product, we also find that δ_n has Dirac's δ-function as its limit. In fact, δ-function is *defined* as the limit of δ_n when it is applied (as convolution) on a Schwartz function.

3 Superposition Principle and Time Evolution

The first axiom of quantum mechanics, superposition principle, claims the same linearity.

Axiom 1. *[Superposition Principle] Physical laws that operate on quantum states shall be linear.*

An implication of superposition principle is how quantum states (precisely, their wave-functions) evolve with time. To give raise to time evolution, we have to add time-dependence to wave-function, which makes a wave-function $f: \mathbb{R}^d \times \mathbb{R} \to \mathbb{C}$, where the second $\mathbb{R}$ represents the time-axis. Then, axiom 1 claims that the equation of time evolution (as a physical law that operates on a quantum state) shall be linear: $\partial f / \partial t = L(f)$ where the operation L is linear.[5] Mathematically, linearity imports a kernel $r: \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{C}$ such that

$$\mathrm{i}\frac{\partial f}{\partial t}(y, t) = \int_{\mathbb{R}^d} \mathrm{d}x\, f(x, t) r(x, y). \tag{1}$$

The right hand side can be seen as a generalization of linear transformation in $\mathbb{R}^n$ like vector-matrix product $\sum_{i=1}^n f_i r_{ij}$. The imaginary i is employed for convenience.

4 Probability Interpretation Implies Hermitianity

Then, the probability interpretation add restriction to the transition rate.

Axiom 2. *[Probability Interpretation] Given a wave-function f of quantum system, the probabilistic density that the system is found at configuration x is given by $|f(x)|^2 = f^*(x) f(x)$.*

Since probabilistic density shall be normalized, axiom 2 implies that, for any wave-function f,

$$\int_{\mathbb{R}^d} \mathrm{d}x\, f^*(x) f(x) = 1. \tag{2}$$

4. This profound construction is found in the lemma 15.1 of Topological Vector Spaces, Distributions and Kernels written by Francois Treves, 1967.

5. Why not $\partial^2 f / \partial t^2 = L(f)$? This may give rise to another axiom that we will know the whole history and future of a wave-function if we know it (or its norm, namely the distribution of particles) at any time t. In short, the evolutionary equation of wave-function is first order on time.

Since this equation holds for all wave-functions, time evolution sustains it too. Namely, for a time-dependent wave-function $f(x,t)$,

$$\int_{\mathbb{R}^d} \mathrm{d}x\, f^*(x,t) f(x,t) = \int_{\mathbb{R}^d} \mathrm{d}x\, f^*(x,t') f(x,t') = 1$$

holds for any $t, t' \in \mathbb{R}$. We are to show what this results in the transition rate r (which determines the time evolution). Equation 2 gives

$$f(y, t+\Delta t) = f(y,t) - \mathrm{i}\Delta t \int_{\mathbb{R}^d} \mathrm{d}x\, f(x,t)\, r(x,y) + o(\Delta t).$$

Plugging into equation 2 for wave-function $f(\cdot, t+\Delta t)$ gives

$$\int_{\mathbb{R}^d} \mathrm{d}y \left[f^*(y,t) + \mathrm{i}\Delta t \int_{\mathcal{X}} \mathrm{d}x\, f^*(x,t)\, r^*(x,y) \right] \left[f(y,t) - \mathrm{i}\Delta t \int_{\mathcal{X}} \mathrm{d}x'\, f(x',t)\, r(x',y) \right] = 1.$$

By inserting equation 2 for wave-function $f(\cdot, t)$ and taking derivative on Δt, it implies

$$\int_{\mathbb{R}^d} \mathrm{d}x \int_{\mathbb{R}^d} \mathrm{d}y\, f^*(x,t) f(y,t)\, [r^*(x,y) - r(y,x)] = 0$$

holds for any f in $\mathcal{S}(\mathbb{R}^d)$, thus

$$r^*(x,y) = r(y,x). \tag{3}$$

Namely, *r is Hermitian*.[6]

5 Path Integral Formalism

We are trying to derive a generic path integral formalism. Given a small $\Delta t > 0$, equation 1 gives

$$f(x', t+\Delta t) = \int_{\mathbb{R}^d} \mathrm{d}x\, f(x,t)[\delta(x'-x) - \mathrm{i}\, r(x,x')\Delta t] + o(\Delta t)$$

We are to convert the $[\cdots]$ part into exponential. To do so, we take the inverse Fourier transform

$$\delta(x'-x) = \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k_\alpha(x'^\alpha - x^\alpha)),$$

and

$$r(x,x') = \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k_\alpha(x'^\alpha - x^\alpha))\, \hat{r}(x,k),$$

in which

$$\hat{r}(x,k) := \int_{\mathbb{R}^d} \mathrm{d}\epsilon \exp(-\mathrm{i}k_\alpha \epsilon^\alpha)\, r(x, x+\epsilon) \tag{4}$$

is the Fourier transform of $\epsilon \mapsto r(x, x+\epsilon)$. Then, the $[\cdots]$ part is converted into exponential by

$$\begin{aligned} [\cdots] &= \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k_\alpha(x'^\alpha - x^\alpha))[1 - \mathrm{i}\hat{r}(x,k)\Delta t] \\ &= \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d} \exp(\mathrm{i}k_\alpha(x'^\alpha - x^\alpha) - \mathrm{i}\hat{r}(x,k)\Delta t) + o(\Delta t) \end{aligned}$$

Plugging back to $f(x', t+\Delta t)$ results in

$$f(x', t+\Delta t) = \int_{\mathbb{R}^d} \mathrm{d}x \int_{\mathbb{R}^d} \frac{\mathrm{d}k}{(2\pi)^d}\, f(x,t) \exp(\mathrm{i}k_\alpha(x'^\alpha - x^\alpha) - \mathrm{i}\hat{r}(x,k)\Delta t) + o(\Delta t)$$

6. A function $f: \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{C}$ is **Hermitian** if $f^*(x,y) = f(y,x)$ for each $x, y \in \mathbb{R}^d$. In the traditional approach of quantum mechanics, we have $r(x,y) = \langle y|\hat{H}|x\rangle$, where $\hat{H}$ is the Hamiltonian operator.

Re-denoting $x_0 := x$, $x_1 := x'$, $k^0 := k$ (for "balancing" indices, we put the "temporal" index of k as superscript, thus the α-component of the k at the 0-th "time-slice" is written by k_α^0),

$$f(x_1, t + \Delta t) = \int_{\mathbb{R}^d} \mathrm{d}x_0 \int_{\mathbb{R}^d} \frac{\mathrm{d}k^0}{(2\pi)^d} \exp(\mathrm{i}k_\alpha^0 (x_1^\alpha - x_0^\alpha) - \mathrm{i}\hat{r}(x_0, k^0)\Delta t) f(x_0, t) + o(\Delta t)$$

Repeating this process N times, we arrive at[7]

$$f(x_N, t + N\Delta t) = \int D(x, k)\, f(x_0, t) \exp(\mathrm{i}S(x, k)) + o(N\Delta t), \tag{5}$$

where the integral

$$\int D(x, k) := \int_{\mathbb{R}^d} \mathrm{d}x_0 \int_{\mathbb{R}^d} \frac{\mathrm{d}k^0}{(2\pi)^d} \cdots \int_{\mathbb{R}^d} \mathrm{d}x_{N-1} \int_{\mathbb{R}^d} \frac{\mathrm{d}k^{N-1}}{(2\pi)^d} \tag{6}$$

and

$$S(x, k) := \sum_{i=0}^{N-1} \Delta t \left[\left(\frac{x_{i+1}^\alpha - x_i^\alpha}{\Delta t} \right) k_\alpha^i - \hat{r}(x_i, k^i) \right]. \tag{7}$$

If we recognize $(x_{i+1} - x_i) / \Delta t$ as the velocity $\dot{x}_i$, then the $S(x, k)$ can be seen as the Legendre transform $\dot{x}\, p - H(x, p)$, in which *k is proportional to momentum p and $\hat{r}(x, k)$ plays the role of Hamiltonian $H(x, p)$*.

6 An Useful Expansion

In this section, we claim an mathematical theorem that is important for continuing the journey. It extends a function in Schwartz space (ensuring Fourier transform), to a generalized function. With this extension, the function can be expanded by a series of generalized functions.

Define the n-th order **moment** (with $n \geqslant 0$) of a function $M_n: \mathcal{S}(\mathbb{R}^d) \to \mathbb{C}$ by

$$M_n^{\alpha_1 \cdots \alpha_n}(f) := \int_{\mathbb{R}^d} \mathrm{d}x\, f(x)\, (x^{\alpha_1} \cdots x^{\alpha_n}). \tag{8}$$

Since $f \in \mathcal{S}(\mathbb{R}^d)$, the moment $M_n(f)$ is finite for any n. Denote $\mathcal{S}_A$ the subspace of Schwartz space in which every function is analytic, termed as **analytic Schwartz space**. Then, for an arbitrary function $\varphi \in \mathcal{S}_A(\mathbb{R}^d)$, Taylor expanding at origin gives

$$\begin{aligned}\int_{\mathbb{R}^d} \mathrm{d}x\, f(x)\, \varphi(x) &= \sum_{n=0}^{+\infty} \frac{1}{n!} \left[\int_{\mathbb{R}^d} \mathrm{d}x\, f(x)\, (x^{\alpha_1} \cdots x^{\alpha_n}) \right] (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \varphi)(0) \\ &= \sum_{n=0}^{+\infty} \frac{1}{n!} M_n^{\alpha_1 \cdots \alpha_n}(f)\, (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \varphi)(0).\end{aligned}$$

On the other hand, because of the identity

$$(\partial_{\alpha_1} \cdots \partial_{\alpha_n} \varphi)(0) = \int_{\mathbb{R}^d} \mathrm{d}x\, \delta(x)\, (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \varphi)(x),$$

integration by parts on the right hand side gives

$$(\partial_{\alpha_1} \cdots \partial_{\alpha_n} \varphi)(0) = (-1)^n \int_{\mathbb{R}^d} \mathrm{d}x\, (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \delta)(x) \varphi(x),$$

where we have omitted the boundary terms since φ is vanishing at boundary. Then, plugging this back, we find

$$\int_{\mathbb{R}^d} \mathrm{d}x\, f(x) \varphi(x) = \int_{\mathbb{R}^d} \mathrm{d}x \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} M_n^{\alpha_1 \cdots \alpha_n}(f) (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \delta)(x) \right] \varphi(x).$$

7. We have to show that the residue is an $o(N\Delta t)$, but this is far from trivial.

Since φ is arbitrary, we finall arrive at

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} M_n^{\alpha_1 \cdots \alpha_n}(f)\, (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \delta)(x). \tag{9}$$

It shall be read as a formal expansion, or an "algorithm" (with which we obtain a valid expression of $\int \mathrm{d}x\, f(x) \varphi(x)$). Only by applying to an analytic Schwartz function can it make sense.

Moments also relate to Fourier transform. We have the Fourier transform

$$\hat{f}(k) = \int_{\mathbb{R}^d} \mathrm{d}x \exp(-\mathrm{i} k_\alpha x^\alpha) f(x).$$

Taking derivatives on k gives

$$\begin{aligned}(\partial^{\alpha_1} \cdots \partial^{\alpha_n} \hat{f})(0) &= \lim_{k \to 0} \int_{\mathbb{R}^d} \mathrm{d}x \exp(-\mathrm{i} k_\alpha x^\alpha) f(x) (-\mathrm{i})^n (x^{\alpha_1} \cdots x^{\alpha_n}) \\ &= (-\mathrm{i})^n \int_{\mathbb{R}^d} \mathrm{d}x\, f(x) (x^{\alpha_1} \cdots x^{\alpha_n}).\end{aligned}$$

Namely, the Taylor expansion coefficient of $\hat{f}$ is $(-\mathrm{i})^n M_n(f)$, and

$$\hat{f}(k) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^n}{n!} M_n^{\alpha_1 \cdots \alpha_n}(f) (k_{\alpha_1} \cdots k_{\alpha_n}). \tag{10}$$

This relates the moments and the Fourier transform of a function in analytic Schwartz space. *We can construct the f by a series of its moments.* For ensuring convergence, we further demand that $\hat{f}$ is analytic, namely $\hat{f} \in \mathcal{S}_A(\mathbb{R}^d)$. (TODO: prove that $\hat{f}_n \in \mathcal{S}_A(\mathbb{R}^d)$ given $f_n(x) = (\delta_n * f)(x)$.)

7 Expansion of Transition Rate

Now we apply the expansion derived in the previous section to transition rate $r(x, y)$. Denote $R_n(x)$ as the n-th order moment of the map $\epsilon \mapsto r(x, x + \epsilon)$, that is (moment is defined by equation 8)

$$R_n^{\alpha_1 \cdots \alpha_n}(x) := \int_{\mathbb{R}^d} \mathrm{d}\epsilon\, (\epsilon^{\alpha_1} \cdots \epsilon^{\alpha_n}) r(x, x + \epsilon). \tag{11}$$

Using equation 9, we directly get

$$r(x, x + \epsilon) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} R_n^{\alpha_1 \cdots \alpha_n}(x)\, (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \delta)(\epsilon). \tag{12}$$

It claims that transition rate r, thus the time evolution of wave-function (equation 1), is completely determined by the moments R_ns.

Plugging equation 12 back to equation 1 gives

$$\mathrm{i} \frac{\partial f}{\partial t}(x, t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_{\mathbb{R}^d} \mathrm{d}w\, f(w, t) R_n^{\alpha_1 \cdots \alpha_n}(w)\, (\partial_{\alpha_1} \cdots \partial_{\alpha_n} \delta)(x - w).$$

Using the parity of $\partial^n \delta$, we change $x - w$ to $w - x$ in $(\partial_{\alpha_1} \cdots \partial_{\alpha_n} \delta)(x - w)$. Then, after integration by parts, we integrate over w, which results in

$$\mathrm{i} \frac{\partial f}{\partial t}(x, t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (\partial_{\alpha_1} \cdots \partial_{\alpha_n}) [R_n^{\alpha_1 \cdots \alpha_n}(x) f(x, t)]. \tag{13}$$

This is a quantum analogy to the Kramers-Moyal expansion in stochastic process.

Interestingly, the Taylor expansion of the "Hamiltonian" $\hat{r}(x, k)$ also relates to the moments R_ns (the $\hat{r}(x, k)$ is defined by equation 4). Indeed, if we Taylor expand $\hat{r}(x, k)$ by k at origin, then the coefficient is

$$\lim_{k \to 0} \frac{\partial}{\partial k_{\alpha_1}} \cdots \frac{\partial}{\partial k_{\alpha_n}} \int_{\mathbb{R}^d} \mathrm{d}\epsilon \exp(-\mathrm{i} k_\alpha \epsilon^\alpha) r(x, x + \epsilon) = (-\mathrm{i})^n \left[\int_{\mathbb{R}^d} \mathrm{d}\epsilon\, (\epsilon^{\alpha_1} \cdots \epsilon^{\alpha_n}) r(x, x + \epsilon) \right].$$

The $[\cdots]$ is recognized as R_n. So, $\hat{r}(x,k)$ can be Taylor expanded as

$$\hat{r}(x,k) = \sum_{n=0}^{\infty} \frac{(-\mathrm{i})^n}{n!} R_n^{\alpha_1\cdots\alpha_n}(x)(k_{\alpha_1}\cdots k_{\alpha_n}). \tag{14}$$

Again, the details of $S(x,k)$ can be completely determined by the moments R_ns.

Comparing with the traditional Hamiltonian $\hat{r}(x,k) = k^2/(2m) + V(x)$, we have all $R_n(x) = 0$ except for $R_0(x) = V(x)$ and $R_2(x) = -1/m$. Equation 12 implies

$$r(x,y) = V(x)\delta(y-x) - \frac{1}{2m}\nabla^2\delta(y-x).$$

Then equation 1 becomes (after integration by parts and omitting the boundary term)

$$\mathrm{i}\frac{\partial f}{\partial t}(x,t) = \int_{\mathbb{R}} \mathrm{d}y\, f(y,t)\left[V(x)\delta(y-x) - \frac{1}{2m}\nabla^2\delta(y-x)\right] = -\frac{1}{2m}\nabla^2 f(x,t) + V(x)f(x,t),$$

which is exactly the Schrödinger equation.

8 Locality Truncates the Moments of Transition Rate

We then introduce the third axiom about locality, and discuss what it will induce.

Axiom 3. *[Locality] Interaction shall be local.*

This implies that the time evolution (equation 1) is local. To make this clear, we consider an example, in which $R_n(x) = c^n$ for some constant c, and set the dimension $d = 1$. Then, the time evolution (equation 13) at $x = 0$ becomes

$$\mathrm{i}\frac{\partial f}{\partial t}(0,t) = \sum_{n=0}^{\infty} \frac{(-c)^n}{n!}\frac{\partial^n f}{\partial x^n}(0,t).$$

The last expression happens to be the Taylor series of $f(x,t)$ at $x = -c$, namely $f(-c,t)$. So, we conclude that $R_n(x) = c^n$ for some constant c implies

$$\mathrm{i}\frac{\partial f}{\partial t}(0,t) = f(-c,t).$$

If we change the value of f at $x = -c$, then the time evolution at $x = 0$ changes accordingly. It means non-locality.

In physics, a local equation generally refers to an operation on f which contains f itself and *finite* number of partial derivatives of f, such as

$$\mathrm{i}\frac{\partial f}{\partial t}(x,t) = \mathcal{L}\left(f(x,t), \frac{\partial f}{\partial x}(x,t), \frac{\partial^2 f}{\partial x^2}(x,t), \ldots, \frac{\partial^n f}{\partial x^n}(x,t)\right),$$

where $\mathcal{L}$ is an analytic function. This is easy to understand because to compute $(\partial^n f/\partial x^n)(0,t)$ using numerical method with difference Δx, only $f(x,t)$ with $x \in \{0, \Delta x, \ldots, n\Delta x\}$ are employed. So, $(\partial f/\partial t)(0,t)$ cannot "perceive" the $f(x,t)$ outside the neighborhood $\{x: |x| \leqslant n\Delta x\}$. Since Δx can be arbitrarily small (but not vanishing), the equation is local.

In the previous discussion, we have shown that, with a cut-off N_{cut} on R_ns such that $R_n = 0$ for any $n > N_{\text{cut}}$, the time evolution is local. And without such a cut-off, we can construct a sequence of R_ns such that the time evolution is not local. Now, we are to prove that, generally, without a cut-off, any sequence of R_ns (such that for any $N > 0$, there are infinite many R_ns that are not vanishing), the time evolution is non-local. This then imports a cut-off on moments.

Before proving, we first declare a property of analytic function. Consider the Taylor expansion of $f \in \mathcal{S}_A(\mathbb{R}, \mathbb{C})$ at origin

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots.$$

The information of f is completely encoded in the infinite sequence of a_ns. This is the result of a theorem which claims that two analytic functions are equal if they agree in any interval (hence we can obtain the Taylor series of the function within the interval). What if we only know a portion of the infinite sequence of a_ns? For example, if we only know the a_0, then only the value of $f(x)$ at $x=0$ is determined. Further, if we also know the a_1, then we can give a good approximation to $f(x)$ in a very tiny neighborhood of $x=0$, since $f(x)=a_0+a_1x+o(x^2)$. Then, if we additionally know the a_2, then the approximation becomes as good as before in a larger neighborhood of $x=0$, since $f(x)=a_0+a_1x+a_2x^2+o(x^3)$ and for keeping the scale of residue, the size of neighborhood can increase a little. This analysis indicates that the more a_ns we know, in a larger neighborhood of $x=0$ can we properly approximate $f(x)$. Our vision becomes border and border if we know more and more a_ns. Until knowing the whole sequence of a_ns, we realize the complete picture of $f(x)$ (based on the previous theorem about analytic function).

It also indicates that, for a sufficient large N, we can keep $f(x)$ (approximately) invariant when we tune the a_ns with $n>N$ while keep the other a_ns invariant. So, based on the equation (plugging $f(x)=\sum_n a_n x^n$ into equation 13, and collecting all a_m terms with $m<n$ into $[\cdots]$).

$$\mathrm{i}\frac{\partial f}{\partial t}(0,t)=\sum_{n=0}^{\infty}\{[\cdots]+(-1)^n R_n(0)a_n\},$$

if there is not a cut-off to the infinite sequence of $R_n(0)$s, we can modify the $(\partial f/\partial t)(0,t)$ by tuning an a_n where n can be arbitrarily large. This, however, will leave the $f(x)$ around $x=0$ invariant. It means that the value of f with x far away from origin can affect the time evolution of f at origin. That is, time evolution is non-local. Hence, there must be a cut-off on the sequence of $R_n(0)$s. This discussion also holds for any value of x other than $x=0$. We conclude that *time evolution is local if and only if there is a positive integer N_{cut} such that $R_n=0$ for any $n>N_{\mathrm{cut}}$*.

9 Hermitianity on the Moments of Transition Rate

Now we study the R_ns. We first investigate what Hermitianity implies for our generic quantum mechanics. Hermitianity implies

$$r^*(x,x+\epsilon)=r(x+\epsilon,x).$$

For simplicity, we temporally abbreviate the notations by $\alpha:=(\alpha_1,\ldots,\alpha_n)$ and $\partial_\alpha^n:=\partial_{\alpha_1}\cdots\partial_{\alpha_n}$.[8] Then, the left hand side is expanded as

$$r^*(x,x+\epsilon)=\sum_{n=0}^{+\infty}\frac{(-1)^n}{n!}(R_n^*)^\alpha(x)(\partial_\alpha^n\delta)(\epsilon),$$

and the right hand side

$$r(x+\epsilon,x)=r(x+\epsilon,(x+\epsilon)-\epsilon)=\sum_{n=0}^{+\infty}\frac{1}{n!}R_n^\alpha(x+\epsilon)(\partial_\alpha^n\delta)(\epsilon),$$

where we have used the relation $(-1)^n(\partial_\alpha^n\delta)(-\epsilon)=(\partial_\alpha^n\delta)(\epsilon)$. Put these together (left hand side minus the right hand side),

$$\sum_{n=0}^{+\infty}\frac{1}{n!}[(-1)^n(R_n^*)^\alpha(x)-R_n^\alpha(x+\epsilon)](\partial_\alpha^n\delta)(\epsilon)=0.$$

Applying on an arbitrary function $\varphi\in\mathcal{S}(\mathbb{R}^d,\mathbb{C})$ (integrating over ϵ), we get (after integration by parts)

$$\sum_{n=0}^{+\infty}\frac{1}{n!}[(R_n^*)^\alpha(x)\partial_\alpha^n\varphi(0)+(-1)^{n+1}\partial_\alpha^n[R_n^\alpha(x)\varphi(0)]]=0,$$

8. It turns out that this is not a convenient abbreviation. Call for optimizations.

or using the original notations,

$$\sum_{n=0}^{+\infty} \frac{1}{n!}[(R_n^*)^{\alpha_1\cdots\alpha_n}(x)(\partial_{\alpha_1}\cdots\partial_{\alpha_n}\varphi)(0) + (-1)^{n+1}(\partial_{\alpha_1}\cdots\partial_{\alpha_n})[R_n^{\alpha_1\cdots\alpha_n}(x)\varphi(0)]] = 0.$$

This is what the Hermitianity of transition rate r induces on the relation between moments R_ns and their complex conjugations R_n^*s.

Expanding this summation term by term, the left hand side becomes

$$\begin{aligned}
&[\![n=0]\!] = (R_0^*)(x)\varphi(0) - R_0^\alpha(x)\varphi(0)\\
&[\![n=1]\!] + (R_1^*)^\alpha(x)\partial_\alpha\varphi(0) + \partial_\alpha R_1^\alpha(x)\varphi(x) + R_1^\alpha(x)\partial_\alpha\varphi(0)\\
&[\![n=2]\!] + \frac{1}{2}(R_2^*)^{\alpha\beta}(x)\partial_\alpha\partial_\beta\varphi(0) - \frac{1}{2}\partial_\alpha\partial_\beta R_2^{\alpha\beta}(x)\varphi(0) - \partial_\alpha R_2^{\alpha\beta}(x)\partial_\beta\varphi(0) - \frac{1}{2}R_2^{\alpha\beta}(x)\partial_\alpha\partial_\beta\varphi(0)\\
&[\![n>2]\!] + \cdots
\end{aligned}$$

Let us examine this temporal result. For simplicity, we set $R_n = 0$ for any $n > 2$. If R_2 is constant, then it implies $R_2 \in \mathbb{R}$ (hint: collecting the term proportional to $\partial_\alpha\partial_\beta\varphi(0)$). It in turn leads to $R_1^*(x) = -R_1(x)$, namely $R_1(x)$ is purely imaginary (this is a surprise). And then, $R_0(x) = R_0^*(x) + \partial_\alpha R_1^\alpha(x)$. If further take R_1 as constant, then we get $R_0: \mathbb{R}^d \to \mathbb{R}$. This is the situation in the traditional Hamiltonian with dissipation.[9] This is the quantum mechanical analogy of Langevin process, where R_n is cut-off at $n = 2$ (such that $R_n = 0$ for any $n > 2$) and R_2 is constant.

Recall that the moments R_ns are cut-off by N_{cut}. The term that proportional to $\partial^{N_{\text{cut}}}\varphi(0)$ is

$$[(R_{N_{\text{cut}}}^*)^{\alpha_1\cdots\alpha_{N_{\text{cut}}}}(x) + (-1)^{N_{\text{cut}}} R_{N_{\text{cut}}}^{\alpha_1\cdots\alpha_{N_{\text{cut}}}}(x)](\partial_{\alpha_1}\cdots\partial_{\alpha_{N_{\text{cut}}}}\varphi)(0) = 0.$$

It implies

$$R_{N_{\text{cut}}}^*(x) = (-1)^{N_{\text{cut}}} R_{N_{\text{cut}}}(x).$$

Namely, $R_{N_{\text{cut}}}(x)$ is real when N_{cut} is even, and imaginary otherwise.

TODO: give a recursive equation that determines the R_ns from $R_{N_{\text{cut}}}$ to R_0.

10 Uncertainty Principle

An **observable** is a real function of the configuration of a quantum system. A corollary of axiom 2 is that, given a wave-function f and an observable $O: \mathbb{R}^d \to \mathbb{R}$, the observed value is the expectation $\mathbb{E}_f[O] := \int \mathrm{d}x\, |f(x)|^2 O(x)$, since $|f(x)|^2$ represents the probability density that the system is found at configuration x. When time is considered, it turns to be

$$\mathbb{E}_f[O(t)] = \int_{\mathbb{R}^d} \mathrm{d}x\, f^*(x,t) f(x,t) O(x). \tag{15}$$

For a single particle system, we use $x(t)$ to denote the position of the particle at time t. Velocity is defined as usual,

$$v(t) := \lim_{\Delta t \to 0} \frac{x(t+\Delta t) - x(t)}{\Delta t}.$$

Both position and velocity are observables. Further, we can compute the variance of an observable O as an expectation $\mathrm{Var}_f[O(t)] := \mathbb{E}_f[(O - \mathbb{E}_f[O(t)])^2]$. Then, uncertainty principle claims the relation between the variances of position and of velocity.

Axiom 4. *[Uncertainty Principle of Single Particle] In a single particle quantum system, given a wave-function f, the variances of position and of velocity have the relation*

$$\Delta x \Delta v := \sqrt{\mathrm{Var}_f[x(t)]\, \mathrm{Var}_f[v(t)]} \sim \hbar/(2m),$$

where m is the mass of the particle, and $\hbar$ is the reduced Planck's constant.

9. If $R_1 \neq 0$, such that $\hat{r}(x,k) = k^2/(2m) + \mu k + V(x)$ for some constant μ, then the time-reversal symmetry breaks. Indeed, μk serves as a friction term. But we usually treat systems with dissipation as incomplete.

10.1 Implication on Moments

In this section, we are to show how uncertainty principle restricts further to the transition rate. We first deal with the situation where dimension $d=1$. Since axiom 4 holds for all wave-functions of a single particle, we choose f to be Gaussian at $t=0$, that is

$$f(x,0)=(2\pi\sigma^2)^{-1/4}\exp\left(-\frac{x^2}{4\sigma^2}\right).$$

The coefficient guarantees the normalization of f. By axiom 2, we have

$$\mathbb{E}_f[x(0)]:=\int_{\mathbb{R}}\mathrm{d}x\, f^*(x,0)f(x,0)x=\frac{1}{\sqrt{2\pi\sigma^2}}\int_{\mathbb{R}^d}\mathrm{d}x\exp\left(-\frac{x^2}{2\sigma^2}\right)x=0,$$

then

$$\mathrm{Var}_f[x(0)]:=\int_{\mathbb{R}}\mathrm{d}x\, f^*(x,0)f(x,0)x^2=\frac{1}{\sqrt{2\pi\sigma^2}}\int_{\mathbb{R}^d}\mathrm{d}x\exp\left(-\frac{x^2}{2\sigma^2}\right)x^2=\sigma^2.$$

To evaluate $\mathrm{Var}_f[v(0)]$, we have to use the time evolution to calculate $x(\Delta t)$. By equation 1, we have (for brevity, we omit the subscript cut in N_{cut})

$$f(x,\Delta t)=f(x,0)-\mathrm{i}\Delta t\sum_{n=0}^{N}\frac{(-1)^n}{n!}\partial^n[R_n(x)f(x,0)]-\frac{\mathrm{i}}{2}\Delta t^2\sum_{n=0}^{N}\sum_{n'=0}^{N}\cdots.$$

We have to evaluate up to $o(\Delta t^2)$, so as to give the variance of velocity which scales as $x^2(\Delta t)/\Delta t^2$. But this would be too complicated (the double summation). For simplification, we consider a “free particle” where all R_ns but the R_N are vanishing. This is the situation when σ tends to zero, because there are more σ factors in the denominator if there are more derivatives on f, and as σ tends to zero, the term proportional to $R_N(x)\partial^N[f(x,0)]$ surpasses all the other terms.[10] In this situation, equation 1 reduces to

$$\mathrm{i}\frac{\partial f}{\partial t}(x,t)=\frac{(-1)^N}{N!}R_N(x)\partial^N f(x,t).$$

Hence,

$$\begin{aligned}f(x,\Delta t)&=f(x,0)+\frac{\partial f}{\partial t}(x,0)\Delta t+\frac{1}{2}\frac{\partial^2 f}{\partial t^2}(x,0)\Delta t^2+o(\Delta t^2)\\&=f(x,0)-\mathrm{i}\frac{(-1)^N}{N!}R_N(x)\partial^N f(x,0)\Delta t-\frac{\mathrm{i}}{2}\frac{(-1)^N}{N!}R_N(x)\partial^N\frac{\partial f}{\partial t}(x,0)\Delta t^2+o(\Delta t^2).\end{aligned}$$

Inserting $(\partial f/\partial t)$ again gives

$$f(x,\Delta t)=f(x,0)-\mathrm{i}\frac{(-1)^N}{N!}R_N(x)\partial^N f(x,0)\Delta t-\frac{1}{2}\left[\frac{(-1)^N}{N!}R_N(x)\right]^2\partial^{2N}f(x,0)\Delta t^2+o(\Delta t^2).$$

Now we are to evaluate $\mathrm{Var}_f[v(0)]$. First, we consider $\mathbb{E}_f[v(0)]=\mathbb{E}_f[x(\Delta t)-x(0)]/\Delta t$. The f used for the expectation is $f(x,\Delta t)$ instead of $f(x,0)$. So, it cannot be expanded as $(\mathbb{E}_f[x(\Delta t)]-\mathbb{E}_f[x(0)])/\Delta t$ except for treating $x(0)$ as a constant, the constant $\mathbb{E}_f[x(0)]$ which has been evaluated as zero. Then, we have

$$\mathbb{E}_f[v(0)]=\frac{\mathbb{E}_f[x(\Delta t)]}{\Delta t}=\int_{\mathbb{R}}\mathrm{d}x\, f^*(x,\Delta t)f(x,\Delta t)\frac{x}{\Delta t}$$

Inserting $f(x,\Delta t)$ up to $o(\Delta t)$, we find

$$\begin{aligned}\mathbb{E}_f[v(0)]&=\int_{\mathbb{R}}\mathrm{d}x\left[f(x,0)+\mathrm{i}\frac{(-1)^N}{N!}R_N^*(x)\partial^N f(x,0)\Delta t\right]\left[f(x,0)-\mathrm{i}\frac{(-1)^N}{N!}R_N(x)\partial^N f(x,0)\Delta t\right]\frac{x}{\Delta t}\\&=\mathrm{i}\Delta t\frac{(-1)^N}{N!}\int_{\mathbb{R}}\mathrm{d}x[R_N^*(x)-R_N(x)]\partial^N f(x,0)x+o(\Delta t).\end{aligned}$$

10. In the traditional approach of quantum mechanics, $\sigma\to 0$ indicates that the momentum is large (since momentum is proportional to $\partial/\partial x$), so the term with the highest order of momentum will dominates the time evolution. When the momentum is large enough, the kinetic term dominates Hamiltonian, the potential becomes omittable, making the particle free (namely, unconstrained by potential).

When N is even, $R_N(x)$ is real (see section 9), we find $\mathbb{E}_f[v(0)] = o(1)$. But when N is odd, $R_N(x)$ is imaginary. In this situation, denote $R_N(x) = \mathrm{i}A(x)/2$, we get

$$\mathbb{E}_f[v(0)] = \Delta t \frac{(-1)^N}{N!} \int_{\mathbb{R}} \mathrm{d}x\, A(x) \partial^N f(x,0) x + o(\Delta t),$$

which may not vanish up to $o(\Delta t)$, leading to a violation of parity symmetry, because the velocity has a favored direction even though the distribution (or wave-function) has not.

Then, using the famous formula of variance,

$$\mathrm{Var}_f[v(0)] = \mathbb{E}_f[v^2(0)] - \mathbb{E}_f^2[v(0)].$$

Now focus on $\mathbb{E}_f[v^2(0)]$. Recalling that $x(0)$ is treated as the constant $\mathbb{E}_f[x(0)] = 0$,

$$\mathbb{E}_f[v^2(0)] = \frac{\mathbb{E}_f[(x(\Delta t) - \mathbb{E}_f[x(0)])^2]}{\Delta t^2} = \frac{\mathbb{E}_f[x^2(\Delta t)]}{\Delta t^2}.$$

Hence, since R_N has been real, we have

$$\begin{aligned}\mathbb{E}_f[v^2(0)] &= \int_{\mathbb{R}} \mathrm{d}x\, f^*(x,\Delta t) f(x,\Delta t) \frac{x^2}{\Delta t^2} \\ &= \left(\frac{1}{N!}\right)^2 \int_{\mathbb{R}} \mathrm{d}x [\partial^N f(x,0) \partial^N f(x,0) - \partial^{2N} f(x,0) f(x,0)] R_N^2(x) x^2 + o(1).\end{aligned}$$

Assume that R_N^2 is analytic, so that we can Taylor expand it at origin, as

$$R_N^2(x) = R_N^2(0) + \partial R_N^2(0) x + \frac{1}{2} \partial^2 R_N^2(0) x^2 + \cdots.$$

We arrive at

$$\mathbb{E}_f[v^2(0)] = \sum_{n=0}^{\infty} \left(\frac{1}{N!}\right)^2 \frac{\partial^n R_N^2(0)}{n!} \int_{\mathbb{R}} \mathrm{d}x [\partial^N f(x,0) \partial^N f(x,0) - \partial^{2N} f(x,0) f(x,0)] x^{2+n} + o(1).$$

10.2 When N Is Even

For simplicity, we first consider the case where N is even. Using maxima, we compute the term in $\mathbb{E}_f[v^2(0)]$ for each N and n.[11] *Restricted N to be positive even number, we shall have $R_N(x) \propto x^{N-2}$ so as to satisfy the uncertainty principle 4.* Any other situation depends on σ, which is either vanishing or diverging when σ tends to zero. The specific case is $N = 2$ in which R_N becomes constant. This is the canonical situation in physics.

10.3 When N Is Odd

TODO

10.4 Generalizations

We can generalize the previous analysis to dimension $d > 1$, in which the covariance matrix of f is diagonal, such that all dimensions are independent. Namely,

$$f(x,0) = \prod_{\alpha=1}^{d} (2\pi(\sigma^\alpha)^2)^{-1/4} \exp\left(-\frac{1}{4}\left(\frac{x^\alpha}{\sigma^\alpha}\right)^2\right).$$

The previous analysis, then, is taken on each dimension individually, resulting in exactly the same result for each dimension.

11. First assume `assume(σ > 0)$`. Define `N: 7$ n: 10$` (for instance), `f: (2*%pi*σ^2)^(-1/4) * exp(-x^2 / (4*σ^2))$`, and `integrand: (diff(f, x, N)*diff(f, x, N) - diff(f, x, 2*N)*f) * x^(n+2)$`. Then `at(diff(R2(x), x, n), x=0)/(N!^2)/n! * integrate(integrand, x, -inf, inf);` gives the n-th term in $\mathbb{E}_f[v^2(0)]$.

In the same way, we can generalize it to multiple particle systems, but it needs an extended uncertainty principle that involves many particles. An extension that can be experimentally examined is:

Axiom 5. *[Uncertainty Principle of Many Particles?] In a multiple particle quantum system, given a wave-function f such that $\mathrm{Var}_f[x_i(0)]$ is small enough for each i-th particle, the variances of position and of velocity have the relation*

$$\Delta x_i \Delta v_i := \sqrt{\mathrm{Var}_f[x_i(t)]\,\mathrm{Var}_f[v_i(t)]} \sim \hbar/(2m_i),$$

where m_i is the mass of the i-th particle, and $\hbar$ is the reduced Planck's constant.

The emphasized part deserves an explanation. At time $t=0$ particles are "prepared" (in quantum mechanics, this can be made by measurement) to be localized. It then leads to a great momentum. When the momenta of particles are large enough, the scattering cross-sections will be small so that the interactions between particles are negligible. Then, we get an extension of uncertainty principle from single particle system to that involves many particles. We can examine this effect in experiments, and use it as an axiom. Then, we obtain the same result as before for each particle, except for the differences in mass.

11 Summary

TODO